Data Structures and Algorithms · 21CSC201J

Unit 2 · Session 4 · SLO 2 — Solved Worksheet

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Unit â€“ 2 : Session â€“ 4 : SLO â€“ 2 | Array Implementation of Lists â€” Programming Practice II

I. Sum and Average

• int sum = 0; for (i = 0; i < n; i++) sum += a[i];
• float avg = (float) sum / n;

II. Armstrong Numbers

• for (i = 0; i < n; i++) { int x = a[i], s = 0, t = x;
• while (t) { int d = t % 10; s += d * d * d; t /= 10; }
• if (s == x) printf("%d is Armstrong\n", x); } // e.g. 153, 370, 371, 407

III. Decimal to Octal

• int oct[32], k = 0, x = num;
• while (x) { oct[k++] = x % 8; x /= 8; }
• while (k) printf("%d", oct[--k]); // prints octal digits in reverse

IV. Sum of Even Numbers

• int s = 0; for (i = 0; i < n; i++) if (a[i] % 2 == 0) s += a[i];

V. Search an Element (Linear Search)

• int found = -1; for (i = 0; i < n; i++) if (a[i] == key) { found = i; break; }
• if (found >= 0) printf("Found at index %d", found); else printf("Not found");